

CourtReserve - One-Page App Summary

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What it is

CourtReserve is a full-stack badminton group slot-booking app with calendar booking, activity tracking, comments, participation analytics, and an AI chat tab.

It combines a React frontend with an Express API backed by PostgreSQL via Drizzle ORM.

Who it's for

Primary users are badminton group members and a lightweight group organizer who coordinate daily weekday slot reservations.

What it does

- Member picker for quick book/cancel actions; remembers the last selected member in a browser cookie.
- Weekday booking window in the UI (next 4 weeks) with slot-capacity limits and duplicate-booking prevention.
- Same-day change lock after 9:30 AM IST for booking and cancellation requests.
- Calendar day cards show booked members, slot usage counts, and booking state by date.
- Recent Activity feed shows latest bookings/cancellations with device and browser details.
- Monthly Participation table computes per-member bookings and participation rate, with sortable columns.
- AI Chat tab streams request-stage progress and can show decision summaries and SQL trace details.

How it works

- Client: React 18 + Vite + Wouter + TanStack Query ('client/src').
- API: Express routes for members, bookings, activities, comments, and AI chat ('server/routes.ts').
- Shared contracts: schema and booking policies reused across runtimes ('shared/*.ts').
- Data layer: Drizzle + PostgreSQL with environment-based schema selection ('server/db.ts').
- AI pipeline: scope gate -> SQL generation -> SQL validation -> read-only execution -> answer synthesis ('server/ai/*').
- Data flow: UI actions -> API calls -> route handlers -> storage/DB -> JSON responses -> React Query cache refresh.

How to run (minimal)

1. Install dependencies: 'npm install'.
2. Create '.env.local' with 'DATABASE_URL=...'; add 'GEMINI_API_KEY=...' for AI Chat responses.
3. Apply DB schema: 'npm run db:push'.
4. Start development server: 'npm run dev' (serves on 'http://localhost:5000').
5. Required Node.js version: Not found in repo.